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A MINI PROJECT REPORT

on

DCTM: Diffusion-Based Cyber Threat Modelling for Improved Intrusion Detection Performance

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BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

by

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ABSTRACT

Machine learning-based Network Intrusion Detection Systems (IDS) remain highly vulnerable to adversarial traffic. While frameworks like DEMGAN utilize multi-generator Wasserstein GANs to achieve an average evasion rate of 97.42% on the CICIDS2017 dataset, they face three critical limitations: a restrictive feature set (limited to 10 features via Mutual Information), a GAN-based instability that fails to generalize against linear classifiers like Decision Trees, and severe dataset class imbalance.

In this paper, we propose a Tabular Diffusion-based Adversarial Framework that overcomes these bottlenecks. Our approach introduces three major advancements:

Diffusion-Driven Generation: We replace the unstable GAN-based architecture with a Transformer-based Denoising Diffusion Model. By learning the data distribution through iterative denoising rather than an adversarial game, our framework produces higher-fidelity adversarial samples that effectively deceive both linear and non-linear IDS classifiers.

Imbalance Mitigation: We integrate SMOTE to handle class imbalance prior to the diffusion process, ensuring the model captures the nuances of rare attack vectors.

Optimized Feature Expansion: We expand the feature set up to 20 attributes through a hybrid analysis of Mutual Information, SHAP methods to identify the most potent feature subset for evasion.

Evaluations on the CICIDS2017 and CICIDS2018 datasets demonstrate that our diffusion-based approach provides more stable and effective adversarial generation. Performance assessment is extended beyond evasion rates to include F1-score, Precision, and Recall, offering a comprehensive and credible benchmark for next-generation IDS resilience.

Keywords: Adversarial Examples, Network Intrusion Detection System, Tabular Diffusion, Transformer, Feature Selection, SMOTE, Class Imbalance, Evasion Attack.

Project Guide

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